

Suggested Teaching Guidelines for
Project
PGCP-AC, August 2026

Duration: 180 Hours: 90 Lab Hours + 90 Self learning Hours

Objective: To design, develop, and deploy a multi-tier, full-stack enterprise web application by seamlessly integrating advanced Java/.NET, Database, and modern web UI frameworks following Agile Software engineering standards.

Prerequisites: Completion of the basic modules on Programming and Database to start Phase I of the Project.

Evaluation method: Grading based on the combined marks obtained in the evaluations of all the 3 phases.

Weightage: Phase I – 10%, Phase II – 20%, Phase III – 20% Phase IV – 50%
(Mid Evaluation 30% + Final Evaluation 70%)

Final Project Schedule

The Project module is divided in four phases.

Phase I: Database Schema & SRS

Schedule: Executed immediately after Database Technologies is completed.

Tasks: Identify the project topic, draft the core requirements (SRS), and design the ER diagram/relational database schema.

Deliverable: Approved SRS document and functional SQL database structure.

Phase II: Frontend Development

Schedule: Executed during and immediately after Web Programming Technologies (WPT).

Tasks: Design the UI/UX layout, wireframes, and code the responsive client-side pages using HTML, CSS, Bootstrap, and JavaScript/React.

Deliverable: Working static frontend prototype with form validations.

Phase III: Backend Integration

Schedule: Executed during Web-based Java Programming and MS.NET Technologies.

Tasks: Create REST APIs, implement business logic, and connect the backend code to the database created in Phase 1 using Spring Boot or .NET Core.

Deliverable: Fully functional backend integrated with the frontend UI.

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Phase IV: Deployment & Final Review

Schedule: This final phase will be executed during the last 15 days of the course. A mid evaluation at the middle of the project development, and a final evaluation at the end of the project will be done.

Tasks: Perform mid-project evaluation, fix bugs, conduct final testing, package the application, and host/deploy it (locally or on cloud platforms).

Deliverable: Live deployed full-stack software system and final project report.